

14.1: Solving Problems

Statistics are numerical properties of a sample & is used to solve real world problems.

1: What problem needs solving? What data needs collecting?
2: How will the data be collected.
3: How will the data be processed & represented
4: How will the data be interpreted? What conclusions can be made?

14.2: Sampling

A population is all possible data, and a sample is a representative part of that data

⇒ The potential values that can be sampled is called the sampling frame
⇒ The proportion of values that are actually sampled is called the sampling fraction
⇒ A 100% sample is a census.

A parameter is a numerical property of a population, and a statistic is a numerical property of a sample.
A parameter for the mean of a population is μ and the statistic for the mean of a sample is $\bar{x}$.

It is not always appropriate to use an entire population to conduct statistical analysis. E.g:
⇒ The data set is very large
⇒ To estimate values of parameters of the parent population
⇒ To conduct a hypothesis test
⇒ To obtain information as a part of a pilot study to inform a proposed investigation.

Questions to consider while sampling:
Is the data biased?
Is the data relevant?
Does data collection affect data?
Is the right person collecting data?
Is the sample large enough?
Is the sampling procedure appropriate?

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Sampling Techniques

Random Sampling	• Minimal Bias • Most theory assumes random sample	• Difficult • Still unrepresentative • Chr /time consuming • Not testable
Stratified (Proportional)	• Each stratum rep^{ed}. • Minimal Bias	• + Complicated than RS • Need info of population to reduce bias • Not testable
Systematic	• Convenient is pop. members can be listed	• Members rarely can be listed
Cluster	• Useful if mems in pop. are grouped in clusters (less travel) • More convenient than R/S Sampling	• Less variation in clusters, biased data

Random:

A random sample of size n has properties:

⇒ Every population member has an equal chance of being selected.

⇒ Every sample of size n, has an equal chance of being made.

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Simple Random Sampling: Selection without replacement

Unrestricted Random Sampling: Selection with replacement.

To obtain a Random Sample:

⇒ All members of population assigned a number.

⇒ Random members of the population are selected through this.

⇒ If URS do not ignore repeats when selecting members.

Stratified Sampling:

Subgroups of a population (age, gender, etc.) are called strata

Stratified Sampling ensures all strata are sampled. If the numbers in each strata which are sampled are proportional to the size of their populations, it is proportional stratified sampling.

Random Sampling is used to select the samples from each strata. Provides more accurate results about the population as well as strata

Cluster:

Samples are taken from specific strata. Investigates specific incidences

Quota:

For opinion polls. Selection left to interviewer, but must meet certain conditions (set number from each strata).

Opportunity:

Whoever is available - will be biased

Self Selecting:

Uses volunteers

Systematic Sampling:

Routine Selection from large sets of data. Pick a random value, add another random value repeatedly and use these as sample member numbers.